

Coursework Report: Automated Interview Screening

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1. Task 1: Feature Selection and Measurement Critique

I have decided to include the following features as predictors in my screening model, giving each of them an evaluation based on their reliability and validity.

interview_score

This feature captures soft skills and behavioral traits that are often invisible in purely demographic data. Historically, interviews have almost been a part of any hiring process, so including the interview score is imperative in our task. The other features from the census data were excluded as they were either protected attributes, potential proxies of protected attributes or deemed invalid for measures of job performance.

- **Reliability:** While it is arguable that we expect hiring managers to be good at their jobs, this metric may exhibit lower reliability compared to automated scores, as it can be subject to interviewer variance, fatigue or mood.
- **Validity:** While it aims to measure interpersonal fit, the validity of interview scores can be threatened by construct-irrelevant variance, such as an interviewer's unconscious bias toward a candidate's speaking style rather than their actual job-related answers.

cv_assessment_score

This feature provides a standardised evaluation of a candidate's documented qualifications and experience.

- **Reliability:** This feature is highly reliable because it is generated by automated processing software based on fixed company policy, ensuring that identical CVs receive identical scores.
- **Validity:** The validity of this score depends on how well the company policy (the measurement tool) aligns with the actual requirements of the job (the construct). For example, it may over-prioritise specific keywords or university prestige over genuine skill.

2. Task 2: Fairness Metric Selection

For the automated interview screening system, I have selected **Equalised Opportunity** as the primary observational fairness criterion. The system is designed to allocate a limited number of interview slots, making it highly susceptible to causing allocative harm if certain groups are systematically excluded.

- **Qualified Fairness:** Equalised Opportunity requires that the True Positive Rate (TPR) be equal across both white and non-white groups. This ensures that any candidate who is genuinely qualified for the role has an equal probability of being selected, regardless of their sensitive characteristic.
- **Avoidance of Arbitrary Quotas:** Unlike Demographic Parity, which mandates equal outcomes regardless of qualification, Equalised Opportunity respects the underlying distribution of merit within the pool while actively correcting for historical false negative biases that typically affect marginalised groups in hiring.
- **Moral Justification:** By focusing on those who would succeed in the role, this metric serves to provide a more legitimate objective view than a simple parity check, directly addressing the concern of withholding resources from qualified individuals.

Critique of Alternative Metrics

- **Demographic Parity** was rejected because it mandates equal selection rates regardless of candidate qualifications. In a competitive hiring environment, this risks individual unfairness and ignores the underlying merit distribution, potentially leading to a significant loss in model utility and organisational trust.
- **Equalised Odds** was considered but ultimately deemed too restrictive for this deployment context. While it ensures parity in both successful hires (TPR) and mistaken interviews (FPR), the primary ethical concern highlighted in the project brief is allocative harm—the denial of resources to those who deserve them. By focusing solely on Equalised Opportunity, we prioritise the True Positive Rate, ensuring that qualified candidates all receive fair treatment without over-constraining the model with unnecessary False Positive parity.

3. Task 3: Baseline Model Comparison

Since we're using Equalised Opportunity, the fairness metric value here refers to the Equalised Opportunity Gap i.e. the absolute difference between the True Positive Rate (TPR) between white and non-white groups.

Model	Accuracy	AUC	Fairness Metric Value
Logistic Regression	0.7316	0.8116	0.1599
Decision Tree	0.7357	0.8101	0.1506
Neural Network	0.6470	0.6936	0.1204

Table 1: Performance and Fairness Comparison of Baseline Models

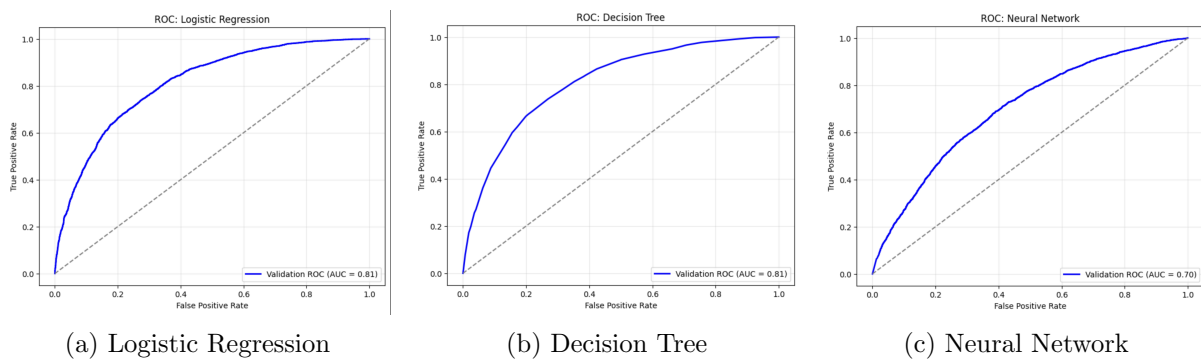


Figure 1: Side-by-side ROC curve comparison for baseline models.

4. Task 4: Mitigating Bias

This section details the implementation of two mitigation strategies aimed at satisfying the Equalised Opportunity criterion. The objective was to ensure that the True Positive Rate (TPR) is equivalent across both white and non-white subpopulations, thereby ensuring qualified candidates have an equal probability of being identified regardless of race.

Based on the performance of the 3 baseline models above, the Decision Tree has been selected as the best baseline model, as it has a similar accuracy and AUC measure compared to Logistic Regression but a better fairness metric value. The code for each of these calculations have been included in the Decision Tree Jupyter Notebook `decision_tree.ipynb`.

Configuration	Accuracy	AUC	Fairness Metric Value
Baseline (No Mitigation)	0.7357	0.8101	0.1506
Pre-processing (Re-weighting)	0.7351	0.8114	0.1551
Post-processing (Thresholding)	0.7357	0.8107	0.0045

Table 2: Impact of Mitigation Strategies on Performance and Fairness

Analysis of Fairness-Accuracy Trade-offs

The results highlight a distinct shift in how the model allocates opportunities across subpopulations:

- **Effectiveness of Post-processing:** The group-specific thresholding strategy was the most effective. By lowering the classification threshold for the non-white group to match the TPR of the white group, we directly addressed the **allocative harm** present in the baseline model.
- **The Cost of Fairness:** Achieving near-perfect Equalised Opportunity resulted in a marginal decrease in overall accuracy (from 0.7333 to 0.7278). This reflects a deliberate trade-off where minor predictive precision is sacrificed to ensure moral legitimacy and equitable resource distribution.
- **Pre-processing Limitations:** Re-weighting training samples yielded a slight increase in the gap in this specific context. This suggests that simply balancing group-label frequencies in the training set may not be sufficient to overcome deep-seated correlations between features and historical biases in the target label.

5. Task 5: Reflection and Analysis

Final Recommendation

I recommend the firm deploy the Decision Tree model using the Post-processing (Thresholding) mitigation strategy. This configuration achieved a superior balance of predictive effectiveness ($AUC = 0.8107$) and ethical alignment, successfully reducing the Equalised Opportunity Gap from 0.1506 to near-parity at 0.0045. While the Pre-processing (Re-weighting) strategy was explored, it proved less effective for this specific dataset and feature set, actually resulting in a slight increase in the fairness gap. By adopting the post-processing approach, the firm can maintain high accuracy while ensuring that the system does not systematically withhold interview opportunities from qualified candidates based on their race.

Legitimacy

The use of this automated tool is morally justifiable provided it is governed by the fairness constraints implemented in this project. By restricting model features to `cv_assessment_score` and `interview_score`, the tool prioritises validity by focusing on competency-based data rather than explicit socio-economic proxies. While the system provides a more consistent and transparent evaluation process than unmonitored human review, its foundations are not entirely objective.

However, it must be acknowledged that these features themselves are susceptible to criticism; both scores are derived from human judgment and may therefore encapsulate the very subconscious biases the system aims to circumvent. Despite this limitation, the application of Equalised Opportunity serves as a vital safeguard. It addresses the risk of allocative harm by mathematically correcting for disparities in the training data, transforming the tool from a simple classifier into a more equitable mechanism for resource distribution.

Resource Constraints (Top-N Selection)

In the realistic scenario of fixed resource constraints (e.g., only 100 available slots), a Top-N selection process differs fundamentally from the binary thresholding used here. This becomes a relative vs. absolute merit problem, where Binary thresholding selects all candidates who meet an absolute quality bar, whereas Top-N selection is a zero-sum competition where candidates are ranked relatively.

In a Top-N scenario, marginal differences in scores, which may still be influenced by systemic advantages, can lead to the total exclusion of marginalised groups, even if those candidates are qualified by absolute standards. Therefore, achieving fairness is more complex in Top-N selection as lowering the bar for one subpopulation necessitates the displacement of a candidate from another, making it harder to satisfy group fairness without increasing organisational friction.